

First Best Benchmark

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Readings: MSPC Chapter 1+2

Introduction

As a first step in our “contract-theory” journey we look at a contractual relationship between an employee and an employer.

- to illustrate the analysis, **terminology**
- define analysis under the first and second best
- to build intuition and illustrate key concepts (contract features/efficiency)
- to illustrate solution procedure/approach under the first best
- in general we will use many different models/environments/notation, but the overall approach is similar.
- lectures will focus on general environment. PS will often give numbers. You have to master both.

Environment

We look at a bilateral relationship (two parties) established through a contract. One of the parties is referred to as the principal (P) the other the agent (A).

- P is responsible for proposing a contract and therefore “moves first”. She offers a contract with at least one wage w .
- A then decides whether to accept the contract or not.

The final result of the relationship, $x \in X = \{x_1, \dots, x_n\}$, depends on the effort, e , provided by the agent, and on the state of nature.

$$p_i(e) = \Pr(x = x_i | e), \text{ for } i = 1, 2, \dots, n$$

Both P and A know the probability distribution of x for each given e .

Assume $p_i(e) > 0$ for all e, i .

Introduction

Central questions:

- Can P through the contract affect how much e A delivers?
- What are the characteristics of the optimal contract?
 - what does the contract look like?
 - how much e is delivered?
 - how much is paid?

e may be:

- observed by P (**symmetric** information (SI))
- unobserved by P (incomplete **asymmetric** information (AI))

We refer to SI as the “first best” situation and AI as the “second best” (after maximizing objective functions etc.).

An analysis of contracts under AI often starts with analysing the contracts under the first best. This provides a useful benchmark to analyse how AI changes the offered contracts and trade in the market.

Central for the problem (under AI) is that there is no outcome realisation that only occurs under certain levels of effort. (Why?)

- I.e. under the second best, P only knows that e.g. success is more likely with high e , but she does not know with certainty whether e was in fact high or low.

Since uncertainty exists it is important what we assume about P 's and A 's attitude towards risk. Attitude towards risk is captured by the utility function. Both parties are expected utility maximizers. P 's Bernoulli utility function:

$$B(x - w)$$

where $B' > 0$, $B'' \leq 0$. What can we say about P if $B'' < 0$?

A 's Bernoulli utility function:

$$U(w, e) = u(w) - v(e)$$

A 's utility function is additively separable, what does this imply?

$u'(w) > 0$, $u''(w) \leq 0$, $v'(e) > 0$, $v''(e) \geq 0$.

$v''(e) \geq 0$ what is the economic argument?

If A does not accept the contract, he gets $\underline{U}$.

Conflict of interest

- P is interested in the result, x . A is not (he is interested in his wage w).
- P is not directly interested in effort e . A is interested in e .
- But P is indirectly interested in effort e because higher e makes better results more likely - P likes higher x .
 - Vi har en interessekonflikt.
 - Under AI, P does not directly observe e .
 - If there were no conflict of interest, A would choose the level of e that P also prefers regardless of complete, incomplete or asymmetric information.

Contract

A contract specifies the terms of trade (verifiable in court).

“A contract is a document that specifies the obligations of the participants, and the transfers that must be made under different contingencies. In order for a contract to be valid, both the contingencies and the terms of the contract must be verifiable”.

Contingencies: Must be observable, verifiable

- under SI: x , contract can also specify e
- under AI: x

Terms of the contract = payment; observable, verifiable

- We proceed by looking at the contractual relationship which arises under symmetric (but not necessarily complete) information.
- We compare this outcome to the contracts which arise under asymmetric information.
- The comparison between the first best and second best allows us to assess the impact of informational asymmetries.
 - First best: e is observable to P (P can simply condition on e)
 - Second best: e is unobservable to P (P cannot condition on e)

Formulation of problem:

- How do we set this up?
- Who are the players?
- Who does what?
- What determines what they do?
- Infinite game tree (one node for each combination of an element in contract space and action space) ... instead of a lot of drawing and guessing we use math....!!
- Pause and think about how to set up this game.

First best problem

Under the **first best**, P can simply choose a level of effort and a wage (potentially varying across states) while securing agent accepts.

Therefore, the first best solution is defined as P maximising expected utility (if risk neutral: expected profits) subject to a participation constraint (PC)/individual rationality (IR) constraint.

- In principle the solution also needs to characterise what happens if the agent chooses an effort level different from the one P demands in the contract (under FB). For now we simply just assume that the fine from this deviation is large enough to secure that A will not do this. (Remember this is only possible since effort is observable and hence verifiable).

Maximization problem - first best

$$\begin{aligned} \max_{e, \{w(x_i)\}_{i=1, \dots, n}} \quad & \sum_{i=1}^n p_i(e) B(x_i - w(x_i)) \\ \text{s.t.} \quad & \sum_{i=1}^n p_i(e) u(w(x_i)) - v(e) \geq \underline{U} \end{aligned} \quad (\text{PC})$$

- Why this setup?
- What does the constraint secure?
- The constraint keeps the agent's expected **utility** from participating in the “game” at least as great as the outside option.

Note that

$$\sum_{i=1}^n p_i(e) u(w(x_i)) - v(e) = \sum_{i=1}^n p_i(e) u(w(x_i)) - v(e) \sum_{i=1}^n p_i(e).$$

- The full solution should also characterise what happens if the agent chooses an effort level different from the one P demands in the contract (under FB).
- For now we simply just assume that the fine from this deviation is large enough to secure that A will not do this. (Remember this is only possible since effort is observable and hence verifiable).

For a **fixed level of effort**, say e^O , the problem is concave in w and hence “well-behaved” (see also MSPC).

Lagrangian:

$$\mathcal{L} = \sum_{i=1}^n p_i(e^O)B(x_i - w(x_i)) + \lambda \left[\sum_{i=1}^n p_i(e^O)u(w(x_i)) - v(e^O) - \underline{U} \right]$$

By the arguments below, the Lagrangian is concave in wages:

- Objective function is concave: $\sum_{i=1}^n p_i(e^O)B(x_i - w(x_i))$
 - $B(\cdot)$ is concave in wages
 - a weighted sum of concave functions is also concave
- Constraint is concave: $\sum_{i=1}^n p_i(e^O)u(w(x_i)) - v(e^O) - \underline{U} \geq 0$

Note: The problem is not necessarily concave in effort (which is why we fixed the level of effort above).

- This makes the solution procedure more complicated (the FOC conditions are only necessary, not sufficient) (Use solution procedure called the “first order approach”)
- Often we will instead be working with restrictions on $p_i(\cdot)$ or assume a finite number of (e.g. two) levels of effort.
- In the latter case, the optimum is found in a final comparison of the objective under different levels of effort.

The FOC associated with $w(x_i)$ is (assuming that $\lambda^O > 0$, we show that this is indeed the case below), for all $i \in \{1, \dots, n\}$,

$$\frac{\partial \mathcal{L}}{\partial w(x_i)} = -p_i(e^O)B'(x_i - w^O(x_i)) + \lambda^O p_i(e^O)u'(w^O(x_i)) = 0 \quad (1)$$

or equivalently

$$\lambda^O = \frac{B'(x_i - w^O(x_i))}{u'(w^O(x_i))} \text{ for all } i \in \{1, \dots, n\}, \quad (2)$$

Since it is not the case at any point that $u'(\cdot) = \infty$ or $B'(\cdot) = 0$, it must be the case that $\lambda^O > 0$. Hence the FOC (1) is valid.

$\lambda^O > 0$ means that the PC binds.

- If $\lambda^O = 0$, the constraint is not binding, i.e. the constraint is not affecting the solution. The value of λ^O is the value of relaxing the constraint, i.e. in this case of lowering $\underline{U}$ (making it easier for P to get A to accept a contract).

The solution to the above problem is Pareto optimal, or efficient.
Why?

Two states of nature, i.e. $x_i \in \{x_1, x_2\}$

Case 1:

If P is **risk neutral** and A is risk averse (2) implies that

$$\frac{1}{u'(w^O(x_i))} = \text{constant} \quad (3)$$

Note: By " P is risk neutral" we mean $B'(\cdot)$ is constant.

Hence, when A is risk averse (3) implies that A must get a fixed wage, independent of x (to secure constant marginal utility across contingencies) and P "absorbs" all the risk in the relationship.

Risk aversion implies decreasing marginal utility, hence for $u'(w^O(x_1)) = u'(w^O(x_2))$ we need $w^O(x_1) = w^O(x_2)$.

Using the binding IR constraint we can now solve for the optimal wage given e^O :

$$\sum_{i=1}^2 p_i(e^O) u(w^O) - v(e^O) = \underline{U} \Leftrightarrow w^O = u^{-1}(\underline{U} + v(e^O))$$

- Sidestep:
 - When P is risk neutral we can simply maximize expected profits, since marginal utility is constant (an expected utility representation is unique up to positive affine transformations).
 I.e. we can maximize:

$$\sum_{i=1}^n p_i(e)(x_i - w(x_i)) = \sum_{i=1}^n p_i(e)x_i - \sum_{i=1}^n p_i(e)w(x_i)$$

- i.e. in assignments with a risk neutral principal you can typically focus on maximizing expected profits
- A risk neutral P only cares about the expected value, A also cares about the allocation of the lottery.

Case 2:

If A is **risk neutral**, $u'(\cdot)$ is constant and therefore (2) implies

$$B'(x_i - w^O(x_i)) = \text{constant}$$

Hence if P is risk averse, the optimal contract must satisfy $x_2 - w^O(x_2) = x_1 - w^O(x_1)$. I.e., A fully “insures” P against all risk.

The optimal contract is of the form:

$$w^O(x_i) = x_i - k$$

The constant k is the payment to P which ensures that the IR constraint binds: (Since A is now risk-neutral, WLOG set $u(w) = w$.)

$$\sum_{i=1}^2 p_i(e^O)(x_i - k) - v(e^O) = \underline{U} \Leftrightarrow \sum_{i=1}^2 p_i(e^O)x_i - v(e^O) - \underline{U} = k$$

Case 3:

If **A and P are both risk averse**, the optimal contract shares the risk (the most risk-averse gets the “lowest amount of risk”).

From the FOC (2) we have that

$$\frac{B'(x_i - w^O(x_i))}{u'(w^O(x_i))} = \text{constant}.$$

I.e. regardless of the realisation of x_i , the solution requires the ratio of marginal utilities to be constant.

When both **A** and **P** are risk averse, the optimal contract shares the risk (the most risk-averse gets the “lowest amount of risk”). One way to show this is to differentiate the FOC (1) wrt x_i :

$$-B''(x_i - w^O(x_i)) \left[1 - \frac{dw^O}{dx_i} \right] + \lambda^O u''(w^O(x_i)) \frac{dw^O}{dx_i} = 0$$

Substitute in for

$$\lambda^O = \frac{B'(x_i - w^O(x_i))}{u'(w^O(x_i))}$$

from (2) and divide through with $B'(x_i - w^O(x_i))$:

$$-\frac{B''(x_i - w^O(x_i))}{B'(x_i - w^O(x_i))} \left[1 - \frac{dw^O}{dx_i} \right] + \frac{u''(w^O(x_i))}{u'(w^O(x_i))} \frac{dw^O}{dx_i} = 0,$$

or

$$r_P \left[1 - \frac{dw^O}{dx_i} \right] - r_A \frac{dw^O}{dx_i} = 0,$$

where r_i is the Arrow-Pratt measure of absolute risk-aversion for $i \in \{P, A\}$.

We can now determine how the optimal contract changes with x_i :

$$\frac{dw^O}{dx_i} = \frac{r_P}{r_P + r_A}$$

The more risk averse A is, the less the state of nature influences the wage. A 's wage varies more with the state of nature the more risk averse P is. Intuition?

Note that r_P and r_A generally depend on $x_i - w^O(x_i)$, and therefore the optimal contract may very well be very complicated. In the case of CARA, the optimal contract will be based on linear payments of the form $w^O(x_i) = c + bx_i$.

- Notice that the allocation of risk is optimal. I.e., the party with the highest cost of risk bears lowest amount of risk.
- This is to be expected given our arguments above regarding efficiency. If risk allocation was not optimal, we would expect there to be additional gains from trade, i.e. the solution would not be efficient.
 - i.e. if the more risk averse party carries too much risk, the less risk averse party can take a part of that risk for a fee that the more risk averse party is willing to pay.

Next step

What happens to the benchmark model under asymmetric information.

- Under asymmetric information, P no longer observes the effort of the agent.
- Intuitively this adds an additional constraint to the problem that the principal is facing.
- The additional constraint ensures that A is induced to take the action wanted by the principal.
 - IR/PC: ensures A participates in the contractual relationship
 - Incentive compatibility constraint (IC): ensures that the agent has the right incentives to perform the action wanted by the principal.

Incentive compatibility

For two levels of effort, e^H, e^L , and two states of nature, the IC constraint for implementing high effort e^H is

$$\sum_{i=1}^2 p_i(e^H)u(w(x_i)) - v(e^H) \geq \sum_{i=1}^2 p_i(e^L)u(w(x_i)) - v(e^L). \quad (IC_H)$$

For continuous effort, the principal's problem is

$$\begin{aligned} \max_{e, \{w(x_i)\}_{i=1, \dots, n}} & \sum_{i=1}^n p_i(e) B(x_i - w(x_i)) \\ \text{s.t.} & \sum_{i=1}^n p_i(e) u(w(x_i)) - v(e) \geq \underline{U} \end{aligned} \quad (\text{PC})$$

$$e \in \arg \max_{\hat{e}} \left\{ \sum_{i=1}^n p_i(\hat{e}) u(w(x_i)) - v(\hat{e}) \right\} \quad (\text{IC})$$

As argued above solving this problem is tricky (the problem is not necessarily concave). The problem is essentially that we have two maximization problems nested in each other. For more see 3.4 in MSPC.